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Physics 326 Section 3

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LAB 9&10: Fourier Series

Introduction

This lab is designed to use Fourier analysis to model and analyze waveforms and voices using a variety of techniques, including using the equipment and numerical tools, as well as how the result from models, particularly waveforms, and different equipment, particularly for Fourier spectrum analysis, differ.

Experiment Setup

The experiment uses a function generator, oscilloscope, Stanford Research Systems (SRS) SR760 FFT Spectrum Analyzer, tuning forks of frequency 440 Hz, 293.67 Hz, and 493.88 Hz, and microphone. Function generator, creating signals of a variety of waves, is connected to the oscilloscope, allowing visualization of generated waveforms, as well as SR760 FFT Analyzer, to analyze its frequency components.

Generated square waves and triangular waves and sound waves by three tuning forks and three voices are collected, with an oscilloscope collecting and visualizing amplitudes as time passes and SR760 Analyzer analyzing and visualizing frequency and amplitude. Each measurement was repeated 3 times.

Results and Discussion

To model wave using trigonometric Fourier series, the model $f(t)$, amplitude vs. time, is given by:

$$f(t) = \frac{A_0}{2} + \sum_{n=1}^{\infty} \{A_n \cos(2\pi n v_0 t) + B_n \sin(2\pi n v_0 t)\} \quad (1)$$

Using Euler's formula, the sinusoidal component can be transformed to:

$$\cos(2\pi n v_0 t) = \frac{e^{2\pi n v_0 t} + e^{-2\pi n v_0 t}}{2}, \quad \sin(2\pi n v_0 t) = \frac{e^{2\pi n v_0 t} - e^{-2\pi n v_0 t}}{2} \quad (2)$$

Therefore,

$$\begin{aligned} f(t) &= \frac{A_0}{2} + \frac{1}{2} \sum_{n=1}^{\infty} (A_n (e^{i2\pi n v_0 t} + e^{-i2\pi n v_0 t}) - i B_n (e^{i2\pi n v_0 t} - e^{-i2\pi n v_0 t})) \\ &= \frac{A_0}{2} + \frac{1}{2} \sum_{n=1}^{\infty} (A_n - i B_n) e^{i2\pi n v_0 t} + (A_n + i B_n) e^{-i2\pi n v_0 t} \end{aligned} \quad (3)$$

Given

$$f(t) = \sum_{-\infty}^{\infty} \hat{f}_n \exp(i2\pi n v_0 t) \quad (4)$$

From equation 3 and 4, given $f(t)$ can be written as sum of such terms, for $n \geq 0$, $\hat{f}_n$ can be written

as $\frac{1}{2}(A_n - i B_n)$; for $n < 0$, $\hat{f}_n$ can be written as $\frac{1}{2}(A_{|n|} + i B_{|n|})$. For even functions $f(t)$, given

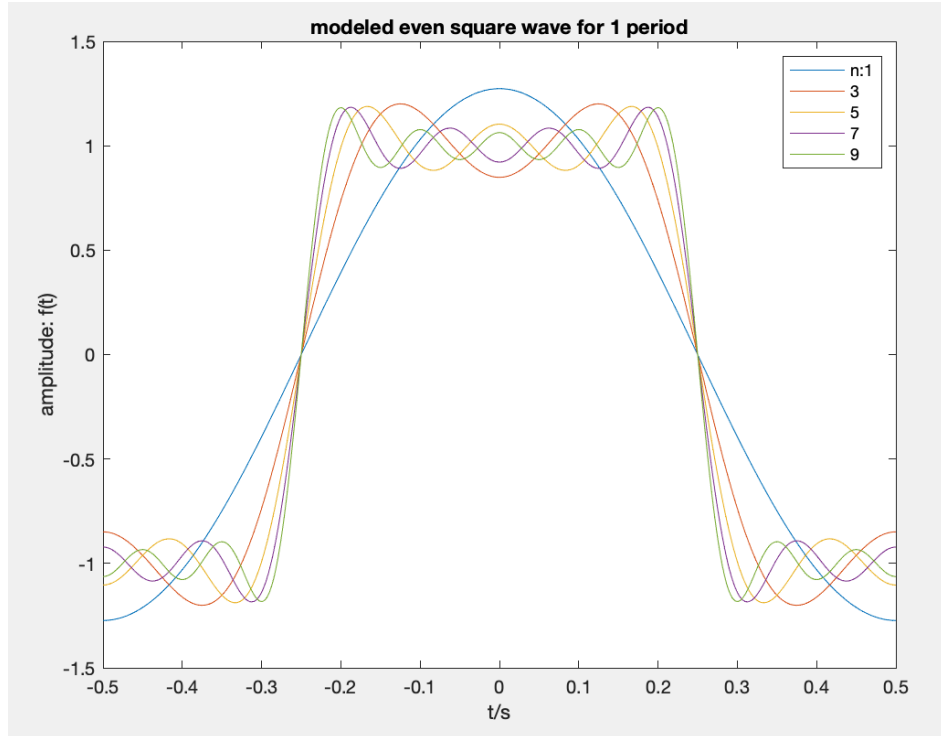
$$B_n = \frac{2}{T} \int_{-T/2}^{T/2} f(t) \sin(2\pi n v_0 t) dt \quad (5)$$

Since the integral is symmetric and $\sin(\theta)$ is odd, the integrand will become odd, which leads B_n , the integral over the symmetric region $[-T/2, T/2]$, to be 0 for all even functions. With similar reasoning, when $f(t)$ is odd, A_n becomes 0 instead, as $\cos(\theta)$ is even.

For a square wave symmetric about y-axis, given the coefficient $A_n = -\frac{4}{n\pi}(-1)^{\frac{n+1}{2}}$, $\widehat{f}_n$

can be written as $A_n/2$. Estimating $f(t)$ for $n_{max}=1, 3, 5, 7, 9$, the models for $T=1$ second are shown in Figure 1:

Figure 1



From the graph, it's observable that $n=9$ gives the wave that's closest to the expected square shape. Such a result fits the expectation, as $f(t)$ is supposed to include n for all integers. It's noticeable that the expected "flat" or the horizontal line part always has the line oscillating to some extent, and such oscillation is smaller for greater n .

For a sawtooth curve with $f(t) = 2t$, with it being odd, only B_n would be nonzero. From equation 5,

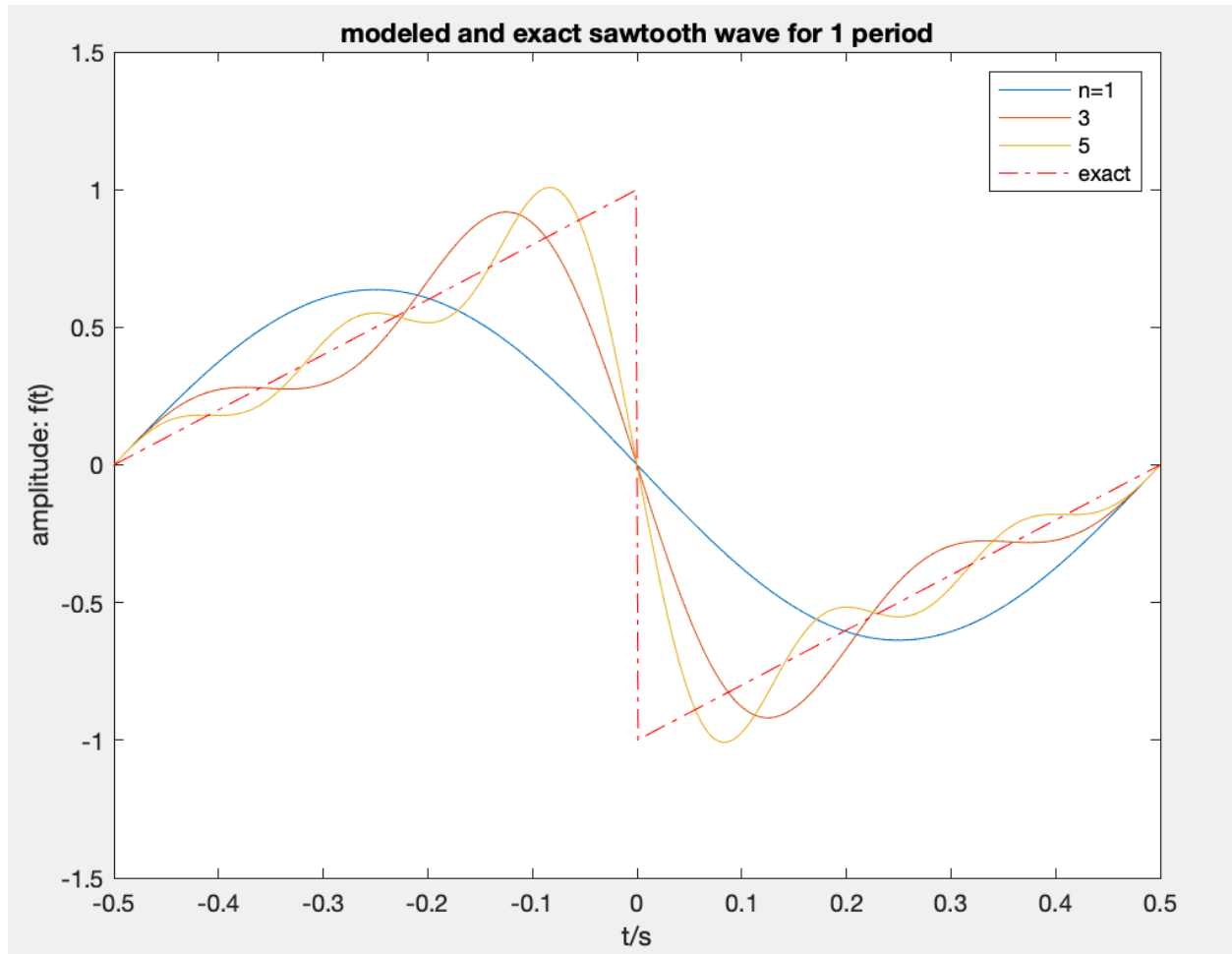
$$B_n = \frac{2}{T} \int_{-T/2}^{T/2} 2t \sin(2\pi n \nu t) dt. \quad (6)$$

Integrating equation 6, we obtain $B_n = -2/\pi n$. Plugging in equation 1, $f(t)$ is obtained:

$$f(t) = -\frac{2}{\pi} \sum_{n=1}^{\infty} \frac{\sin(2\pi n \nu t)}{n} \quad (7)$$

For $n_{max} = 1, 3, 5$, the estimated curve is shown in Figure 2:

Figure 2



It still follows that higher $n_{\max}$ gives a better estimation of the curve. The small oscillations around the “horizon” part, however, still don't disappear.

Using a function generator, square waves are first generated with varying voltages. Modeling with $n=21$, Figure 3 shows modeled and exact values for waveforms, and Figure 4 shows Fourier spectrum obtained from waveform and SR760 Analyzer:

Figure 3

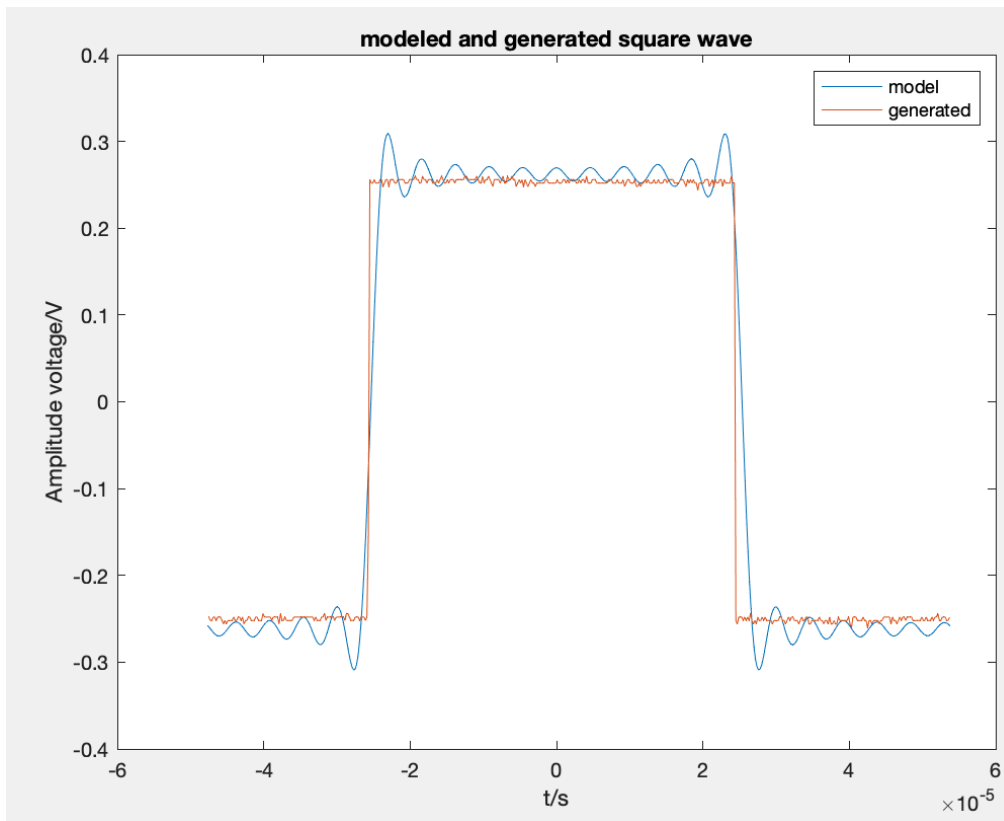
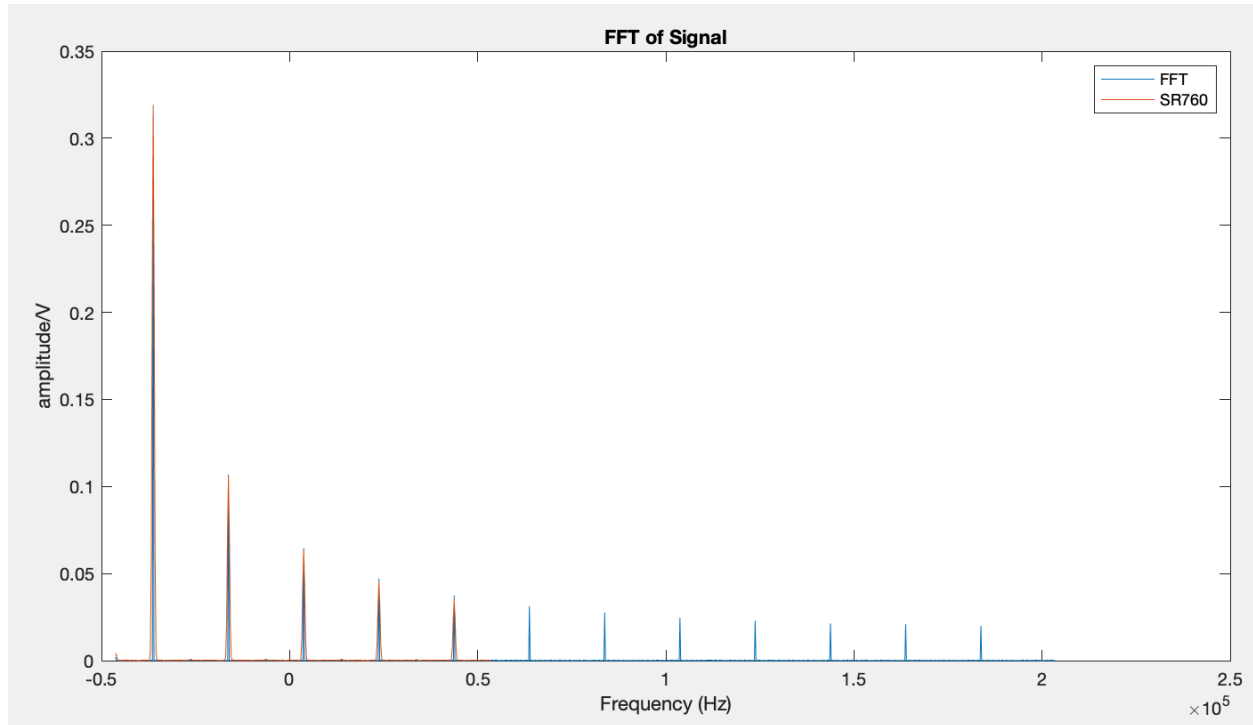


Figure 4



It could be seen that the waveform generated is consistent with the model in the range of data provided by SR760, and the Fourier spectrum from SR760 is overall consistent with the computed one from the waveform, besides that amplitude of peaks does vary, possibly explained by environmental factors. When SR760 is switched to logarithmic view, smaller peaks appear, and the data outside of the peaks has a variance that's more easily observed. With idealistic Fourier series, there shouldn't be peaks at where n is even, as its Fourier coefficient is 0. The observed peaks might come from environmental factors, causing squarewaves to be not perfectly generated, as suggested by that Fourier spectrum synthesized by generated waveforms was also continuous with peaks. The peaks, therefore, should be accounted for by the logarithmic scale that reduces the extreme variances.

The same process was repeated for triangular waves in Figure 5, showing the modeled wave with $n=21$ and generated triangular wave for 1 period, and Figure 6, Fourier spectrum from waveforms and SR760.

Figure 5

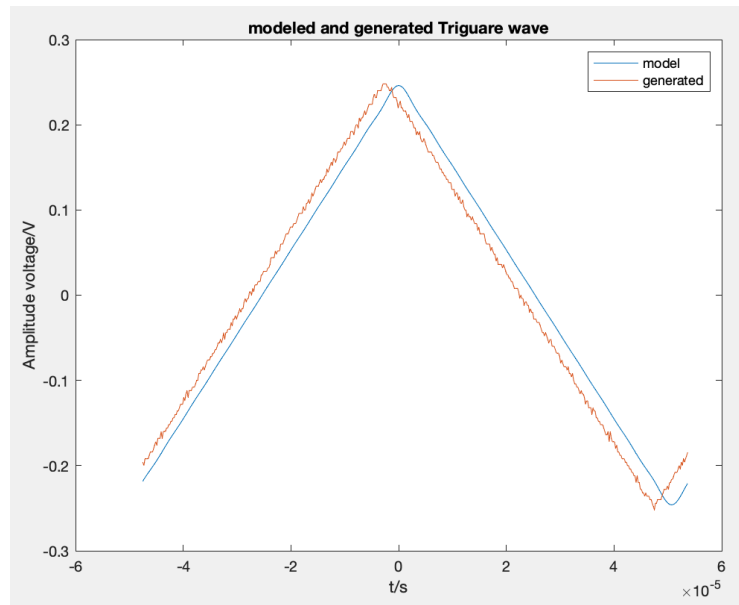
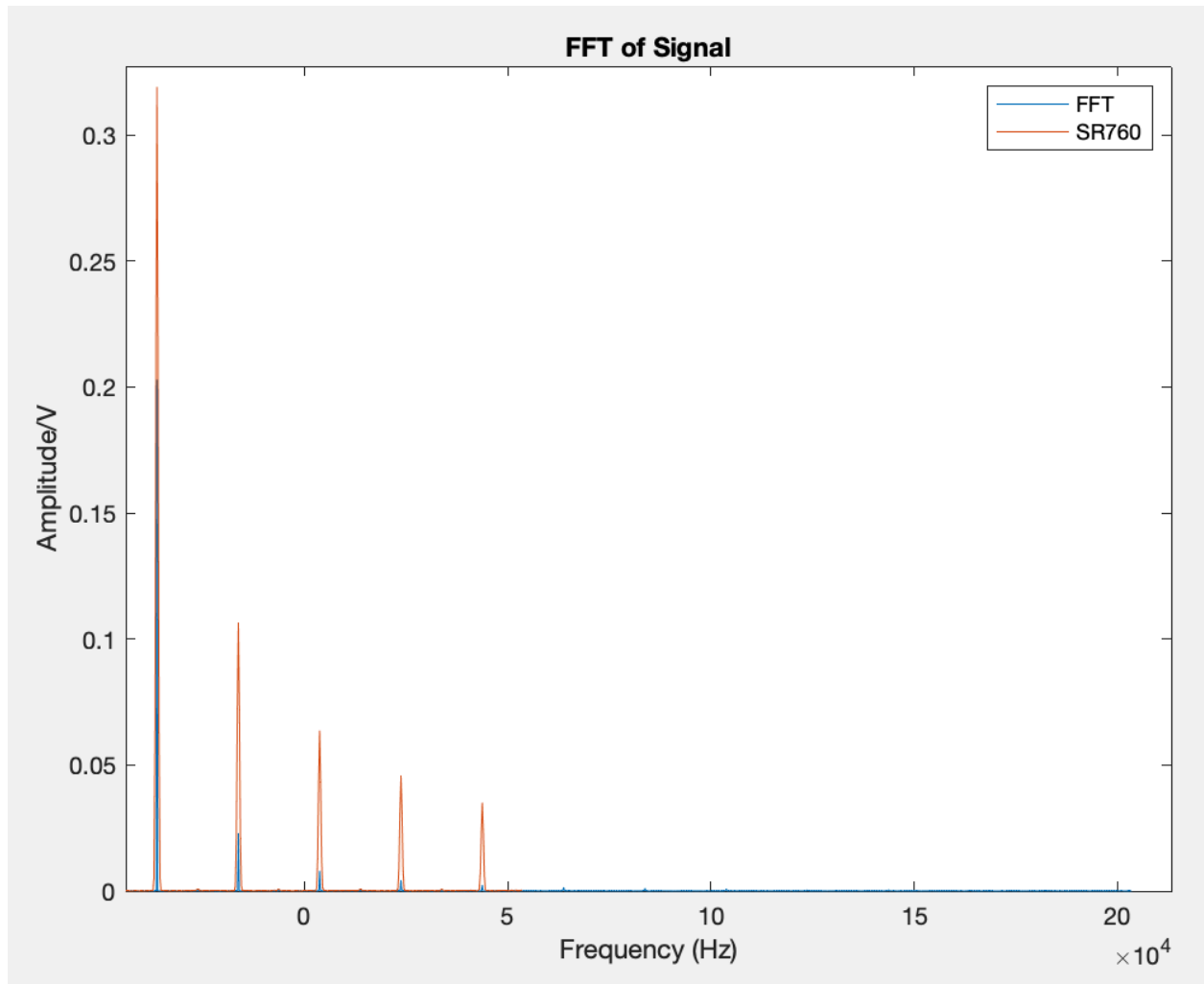


Figure 6



The same consistency was given.

For three tuning forks of frequency 440 Hz, the waveform and Fourier spectrum was collected and analyzed in Figure 7, showing Fourier spectrum of linear scale, and Figure 8, the one in logarithmic scale.

Figure 7

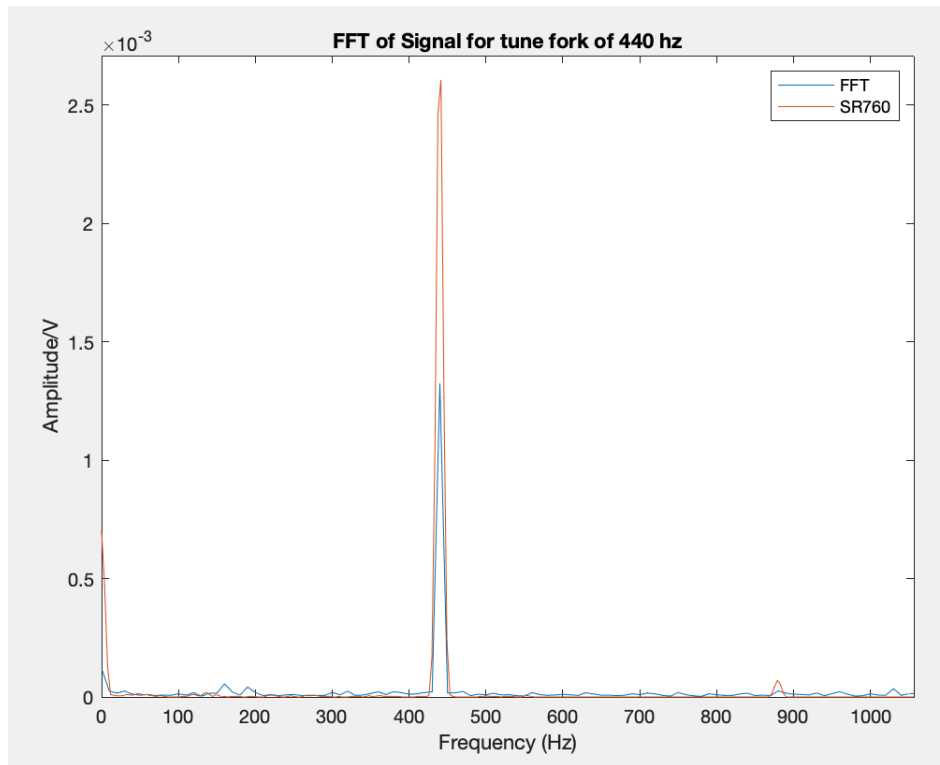
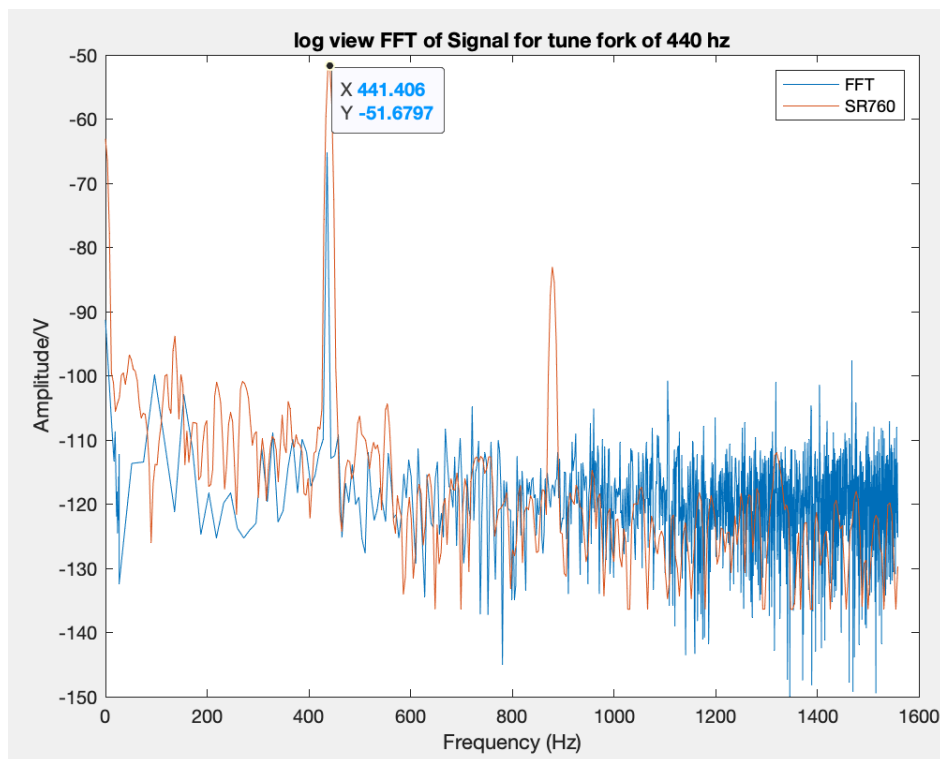


Figure 8



The linear scale shows a consistency between Fourier spectrum computed by data collected from an oscilloscope. As expected by the frequency given by tuning fork's label, there's a peak near 440 Hz. However, it shows a relatively great variance in the logarithmic scale, besides that they both have a significant peak. The secondary large peak from SR760 was not present in the computed spectrum from oscilloscope, and less significant variance is present in the Fourier spectrum computed. It's possibly because SR760 has a better sensitivity of voice, especially when logarithmic scale enlarges negligible voices. Additionally, the way SR760 processes the voice is unknown. When computing the logarithmic scale of data collected from the oscilloscope, the only known way is to take the logarithmic of it and scale it. Combined with its advanced processing, sharper peaks are given by SR760.

Linear views for the remaining tuning forks are displayed in Figure 9 and Figure 10.

Figure 9

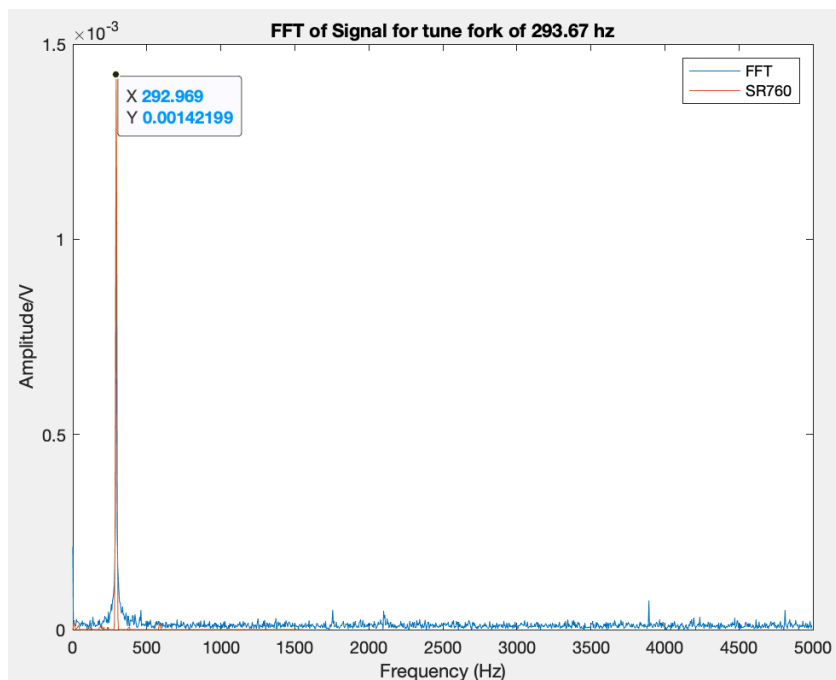
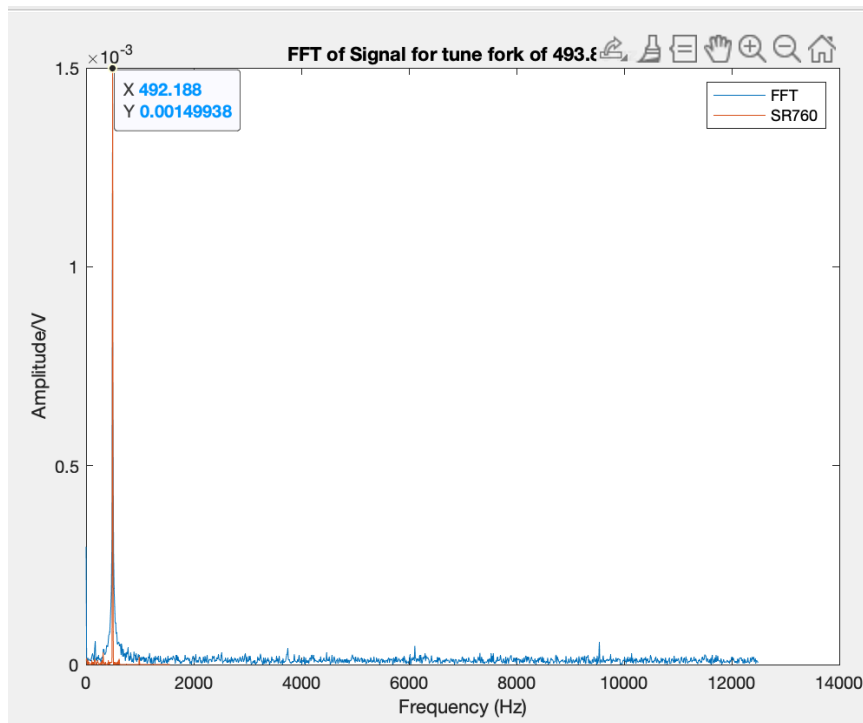


Figure 10



Both of the spectrums show a consistency between positions of the peaks, and peaks do appear at where it's expected. The tuning forks generally show a pure voice, as the spectrum given by SR760 has a larger amplitude of the peaks and smaller amplitudes for where peaks are not present.

Three different voices are presented in Figure 11, 12, and 13. The second voice was sound when speaking, and the remaining voices attempted to be a single tone.

Figure 11

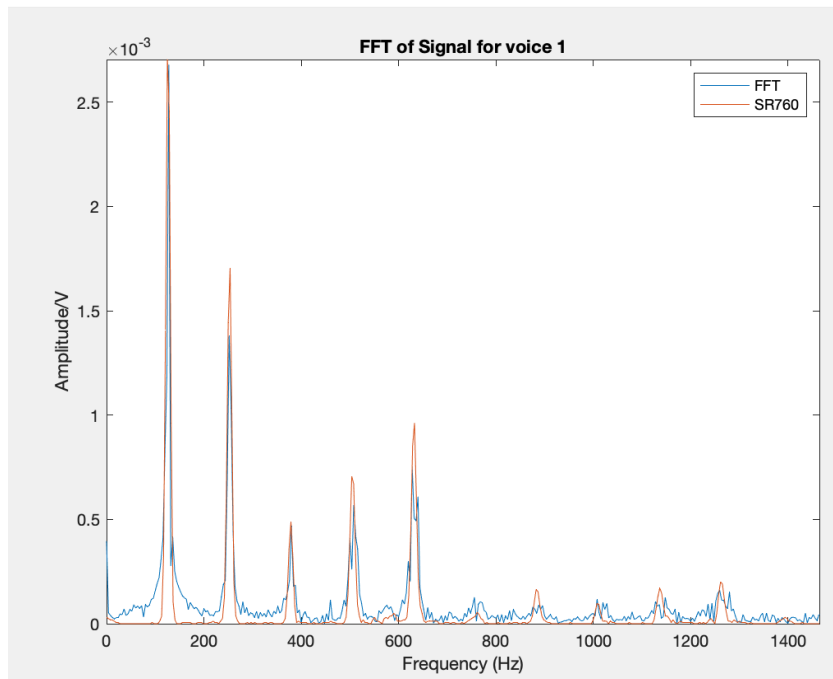


Figure 12

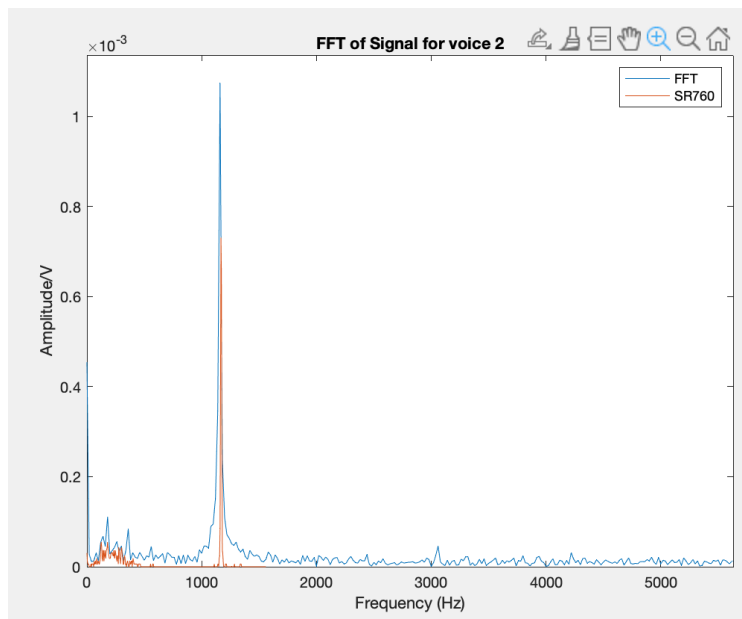
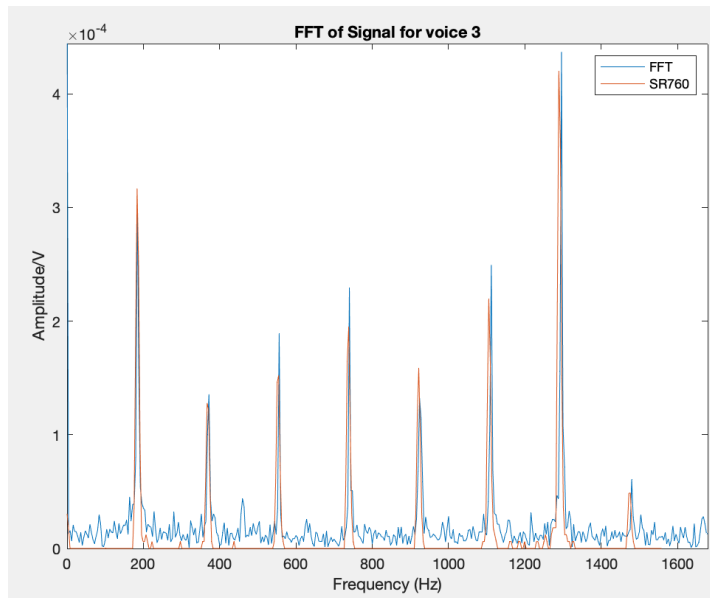
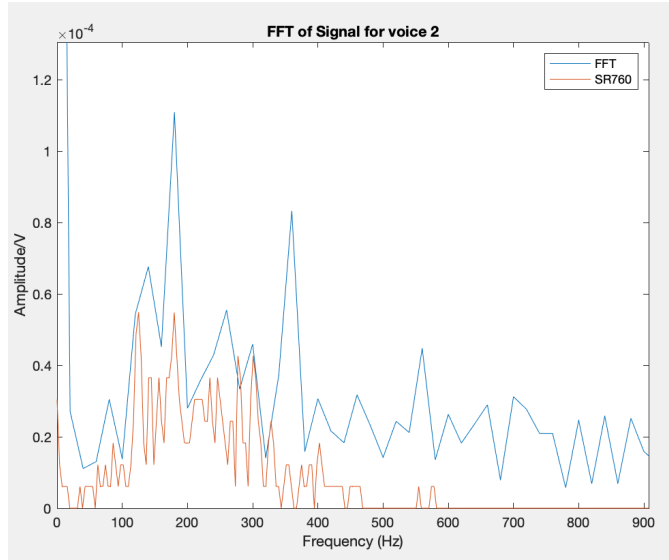


Figure 13



The voices are not as pure as tuning fork, as there are more smaller peaks under the most significant peaks. Such a phenomenon is natural, as the voices are human voices. For voice 1 and voice 3 that are in single tone, it could be seen that peaks appear at approximately the same intervals. For voice 2, the spoken sound, there's a peak over 1000 Hz, indicating some background noise. Figure 14 zooms in the possible range where the spoken voice is.

Figure 14



It suggests a spoken voice. However, as spoken voice has a broader range, greater variance is produced, so the pattern for where peaks appear is less likely to be observed. Additionally, given SR760 gives a greater peak and smaller amplitudes for where peaks are, it's likely that SR760 eliminated the spoken sound to some extent when the tuning fork at the background was sound. Overall, when voices are more pure, peaks' positions for computed results and SR760 will be consistent.

Conclusion

In this lab, waveforms for generated waves and Fourier spectrum from SR760 and computed results from the oscilloscope are analyzed. Fourier series could give a relatively accurate approximation for waveforms. The Fourier spectrum from SR760 seems to give a more pure result, and positions where peaks appear are generally consistent for these two spectrums. The logarithmic scale, however, gives a confusing spectrum, possibly due to the way SR760 processes the voice. To avoid such a difference, better techniques for eliminating background could be applied.